Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Other Protocols in Action

Introduction to FTP and SNMP.Practical (30 min):Set up an FTP server and transfer files between devices.

Task: Create a table comparing FTP and HTTP.

❖ **1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.**

➢ **Aim**

- To install and configure FileZilla Server on a Windows computer, create an FTP user account, and share a folder so that files can be accessed and transferred using the FTP protocol.

➢ **Introduction**

- FTP (File Transfer Protocol) is a standard network protocol used to transfer files between computers over a network. It allows users to upload and download files from an FTP server.

- FileZilla Server is a free FTP server application that allows a computer to act as a file-sharing server. Other devices can connect to it using an FTP client such as FileZilla Client.

➢ **Requirements**

1. Windows computer or laptop.
2. Internet connection.
3. FileZilla Server software.
4. A folder containing files to share.
5. Another device or FileZilla Client for testing the FTP connection.

➢ **Procedure / Steps**

➢ **Step 1: Download FileZilla Server**

➢ Open a web browser.

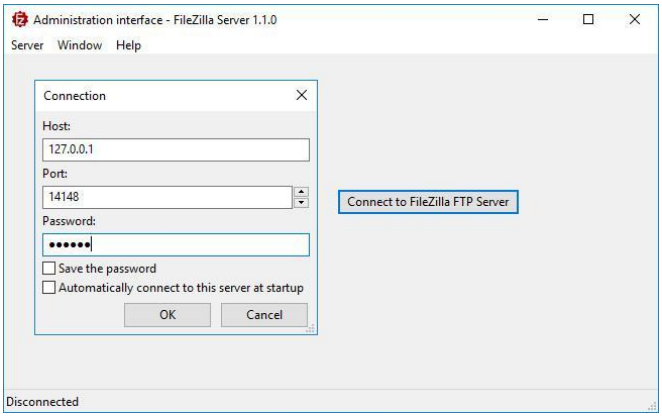
➢ Visit the official website: https://filezilla-project.org/

➢ Download FileZilla Server for Windows.

➢ Run the downloaded installation file.

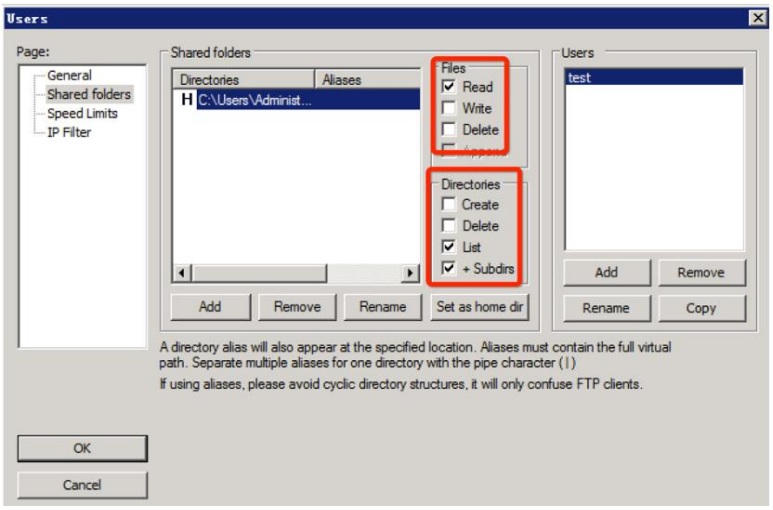


- **Step 2: Install FileZilla Server**
- Accept the license agreement.
- Select the standard installation option.
- Choose the required installation location.
- Configure the server to run as a Windows service.
- Complete the installation process.
- Open the FileZilla Server Administration Interface.
- **Step 3: Connect to the Server**
- Open FileZilla Server Administration Interface.
- Enter the server address as 127.0.0.1 (localhost).
- Enter the administration port configured during installation (commonly 14148).
- Click on Connect.
- The server administration interface will open.

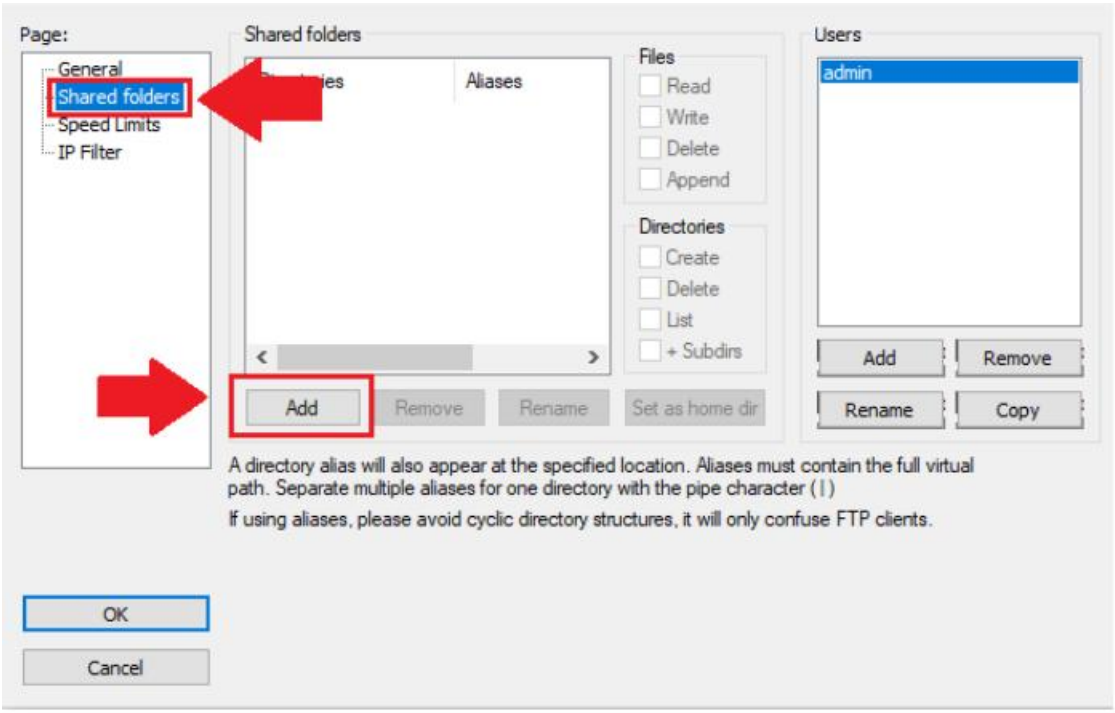


- **Step 4: Create a Shared Folder**
- Create a new folder on your computer, for example, C:\FTP_Shared.
- Open FileZilla Server settings.
- Go to the Users section.

- Select Shared Folders.
- Add the folder C:\FTP_Shared as the shared directory.
- Set it as the Home Directory.
- Give the required permissions, such as Read and Write.



- **Step 5: Create an FTP User Account**
- Open the Users section in FileZilla Server.
- Click on Add to create a new user.
- Enter a username, for example, student.
- Enable password authentication.
- Set a secure password for the user.
- Select the shared folder created earlier.
- Assign the necessary permissions to the user.
- Save the configuration.

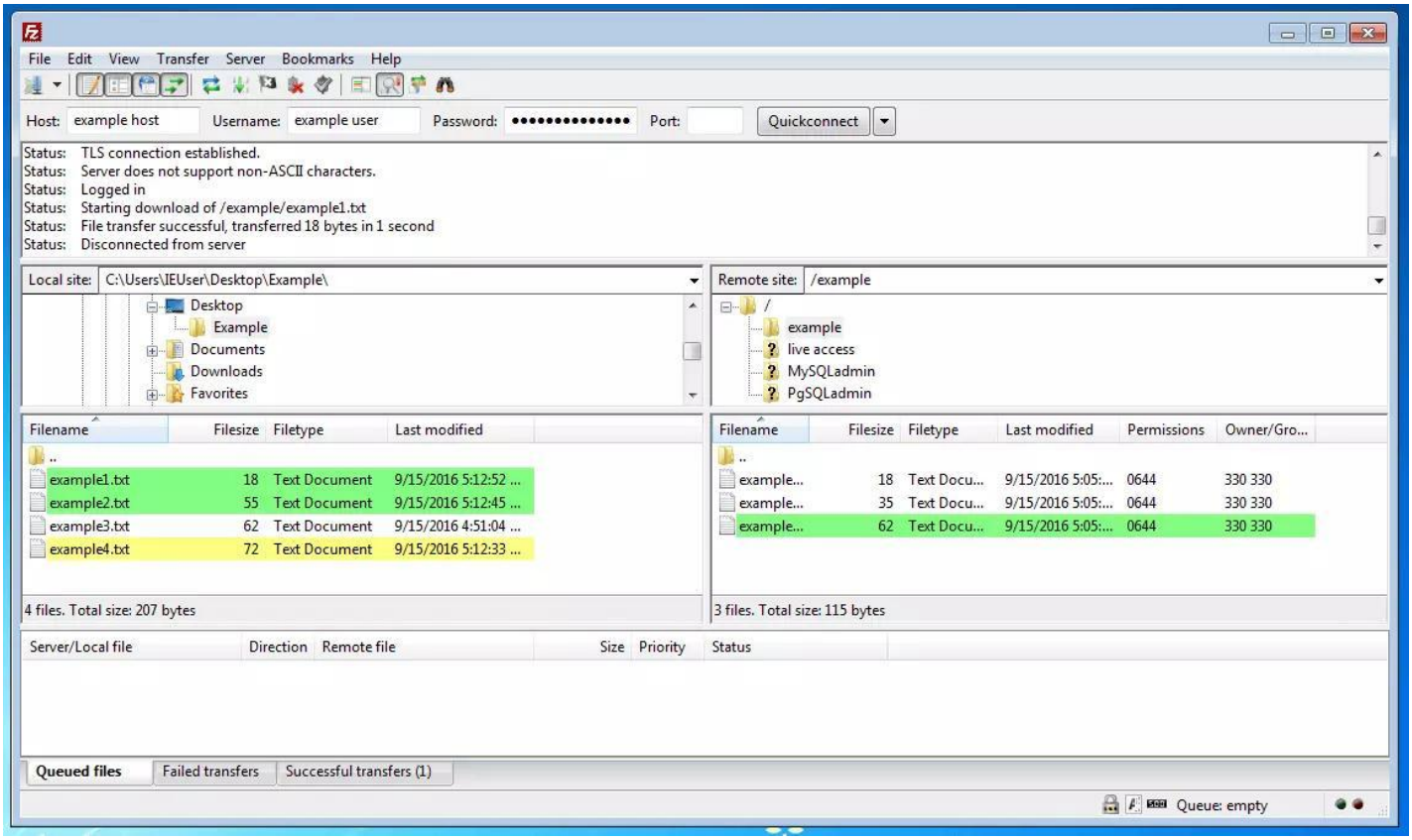


- **Step 6: Connect and Transfer Files**
- Open FileZilla Client on the same computer or another device.
- Enter the following connection details:

Field	Value
Host	Server IP address
Username	student
Password	Your configured password
Port	21 (if configured for FTP)

- Click on Quickconnect.
- After successful connection, the shared folder will appear.

- Select a file from the local computer.
- Drag and drop it into the remote server folder to upload it.
- Download files from the server to the local computer in the same way.



- **Result**
- FileZilla Server was successfully installed and configured on the Windows computer. An FTP user account was created, and a shared folder was configured with the required permissions. Files could be uploaded and downloaded through the FTP connection.

❖ **2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.**

➢ **Aim:**

- To connect to an FTP server using FileZilla Client with the created user credentials and upload an image file from the local computer to the server's shared folder.

➢ **Introduction**

- FileZilla Client is a free FTP client application used to connect to FTP servers and transfer files between computers.
- It allows users to upload files from their local computer to a remote server and download files from the server to their computer.

➢ **Requirements**

1. Windows computer or laptop.
2. FileZilla Client software.
3. FileZilla Server installed and running.
4. FTP username and password.
5. An image file (JPG or PNG format).
6. Server IP address and FTP port number.

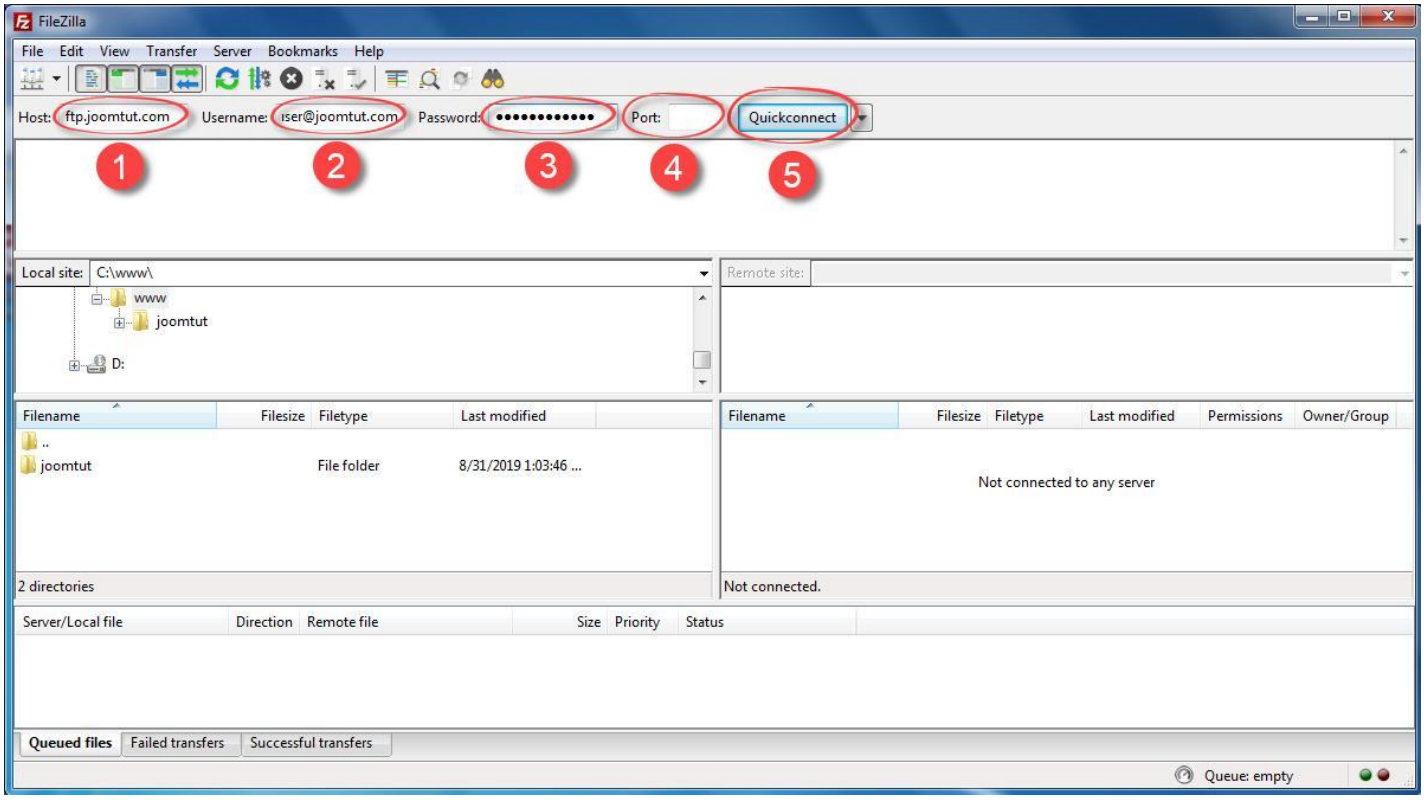
➢ **Procedure / Steps**

➢ **Step 1: Open FileZilla Client**

➢ Open FileZilla Client on your Windows computer.

➢ The main interface will appear.

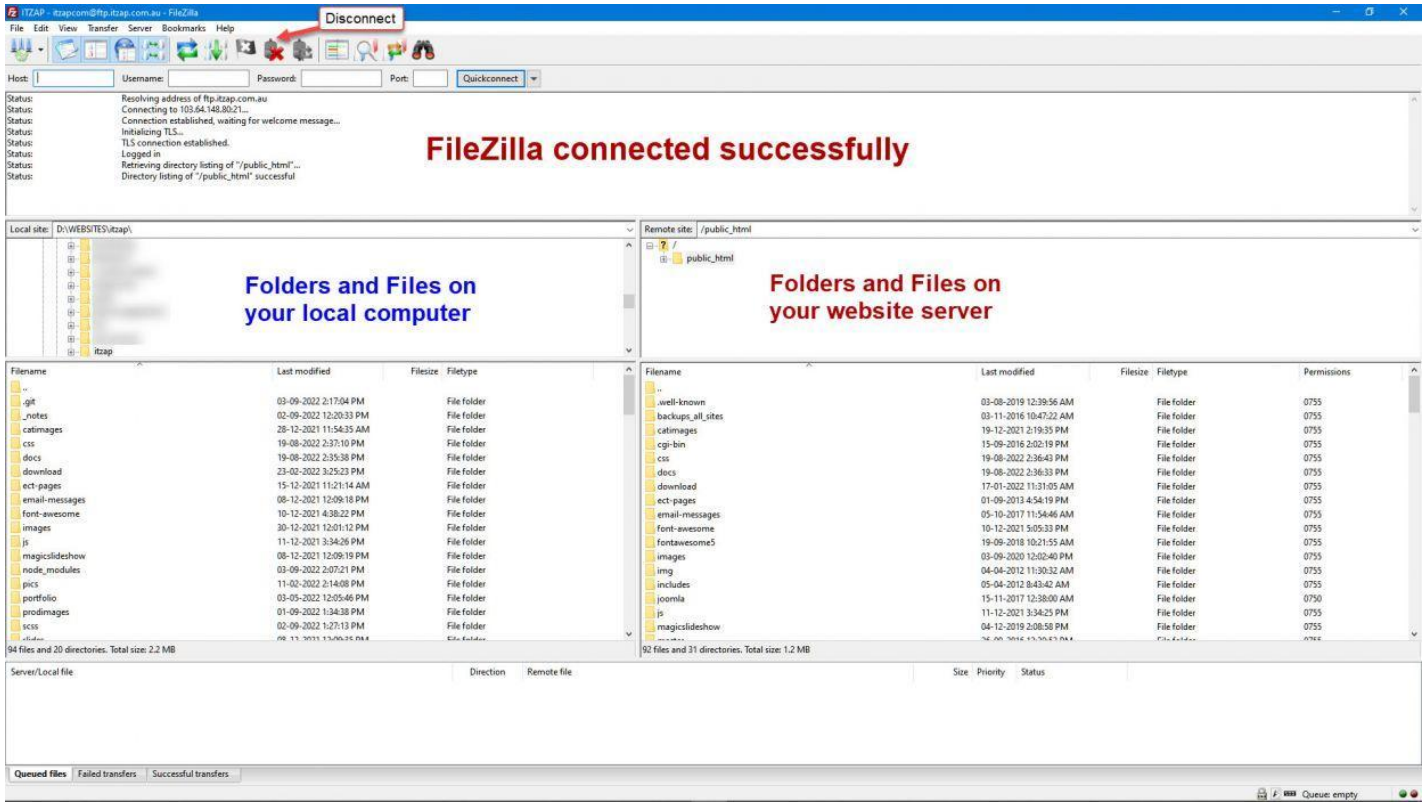
- At the top, you will find Host, Username, Password and Port fields.



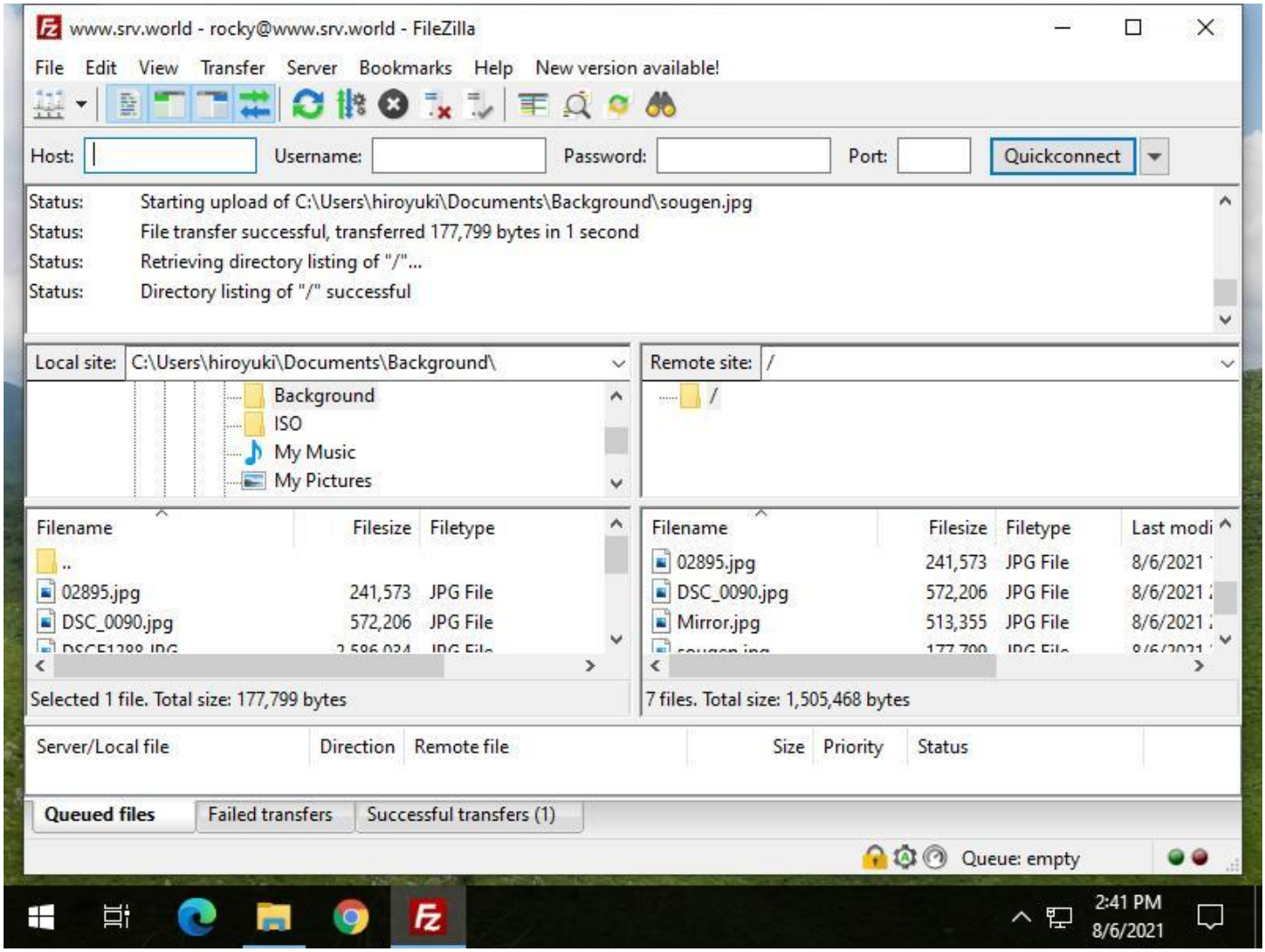
- **Step 2: Enter FTP Server Credentials**
- Enter the FTP server details in the Quickconnect section.

Field	Details
Host	FTP Server IP address
Username	student
Password	Created FTP password
Port	21 (if configured for FTP)

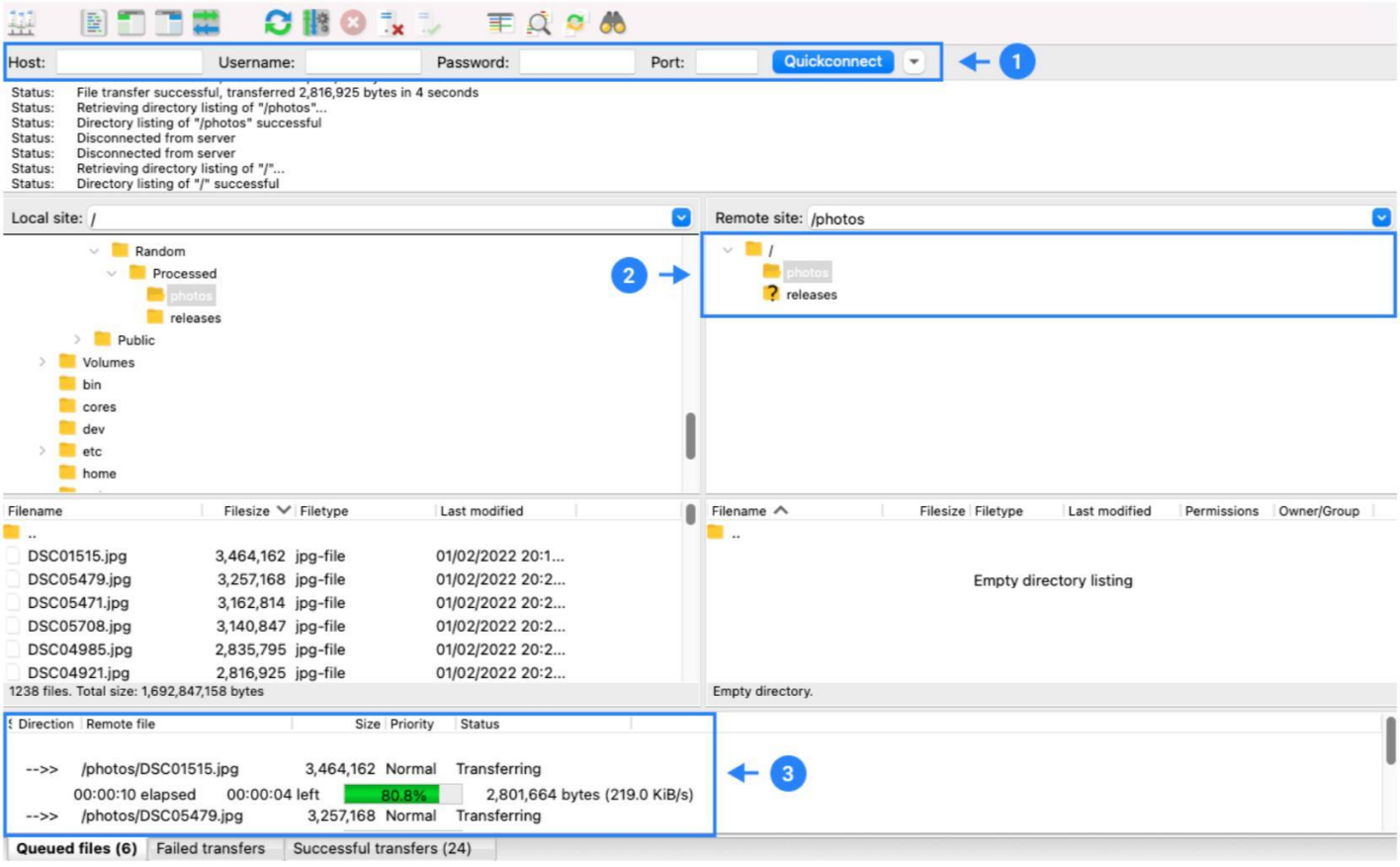
- Enter the correct server IP address.
- Enter the username and password created in FileZilla Server.
- Enter port number 21.
- Click on Quickconnect.
- **Step 3: Establish the FTP Connection**
- FileZilla Client will attempt to connect to the FTP server.
- If the credentials are correct and the server is accessible, the connection will be established.
- A successful connection message will appear in the message log.
- The remote server files will be displayed on the right side.



- **Step 4: Select an Image File**
- On the left side, locate the Local Site section.
- Browse your computer folders.
- Select an image file in JPG or PNG format.
- For example, MyPhoto.jpg.

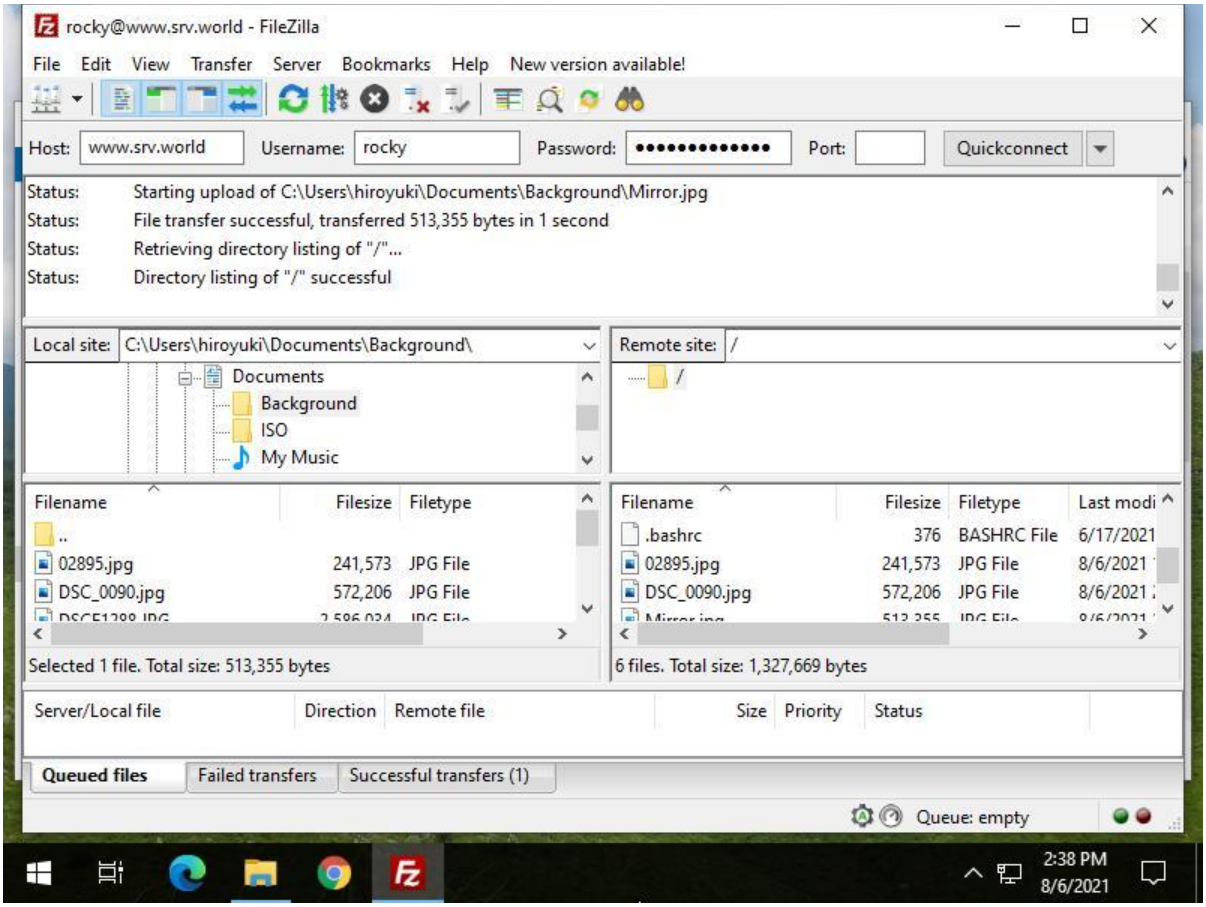


- **Step 5: Upload the Image File**
- On the right side, locate the Remote Site section.
- Open the shared folder configured on the FTP server.
- Select the image file from the Local Site.
- Right-click and select Upload, or drag and drop the file into the Remote Site.
- FileZilla will start transferring the image file to the FTP server.



- **Step 6: Verify the Uploaded File**
- After the transfer is completed, check the Remote Site section.

- The uploaded image file should appear in the shared folder.
- Check the Successful Transfers section to confirm that the file was transferred successfully.



- **Result**
- Successfully connected to the FTP server using FileZilla Client with the created username and password. An image file was uploaded from the local computer to the FTP server's shared folder and verified successfully.

❖ 3. Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

➢ Aim:

- To download an uploaded file from an FTP server to a different folder on the local computer using FileZilla Client and verify that the downloaded file size matches the original file size.

➢ Introduction

- FileZilla Client is an FTP client application used to transfer files between a local computer and a remote FTP server.
- It provides two main sections: Local Site and Remote Site. Users can download files from the remote server to their computer and verify whether the file has been transferred correctly by comparing file sizes.

➢ Requirements

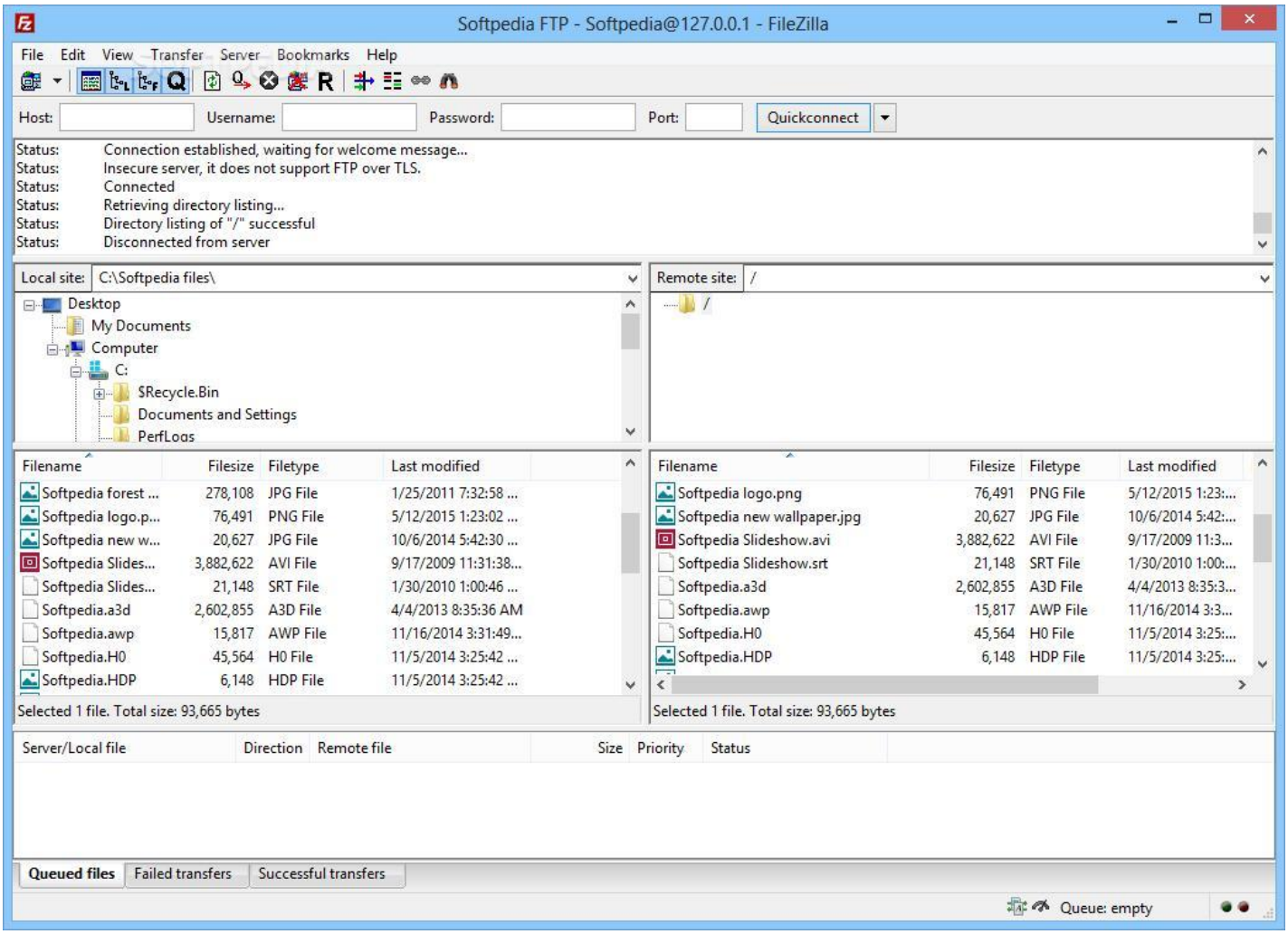
1. Windows computer or laptop.
2. FileZilla Client software.
3. Running FileZilla FTP Server.
4. Valid FTP username and password.
5. Previously uploaded image file.
6. A separate destination folder on the computer.

➢ Procedure / Steps

➢ Step 1: Open FileZilla Client

➢ Open FileZilla Client on your computer.

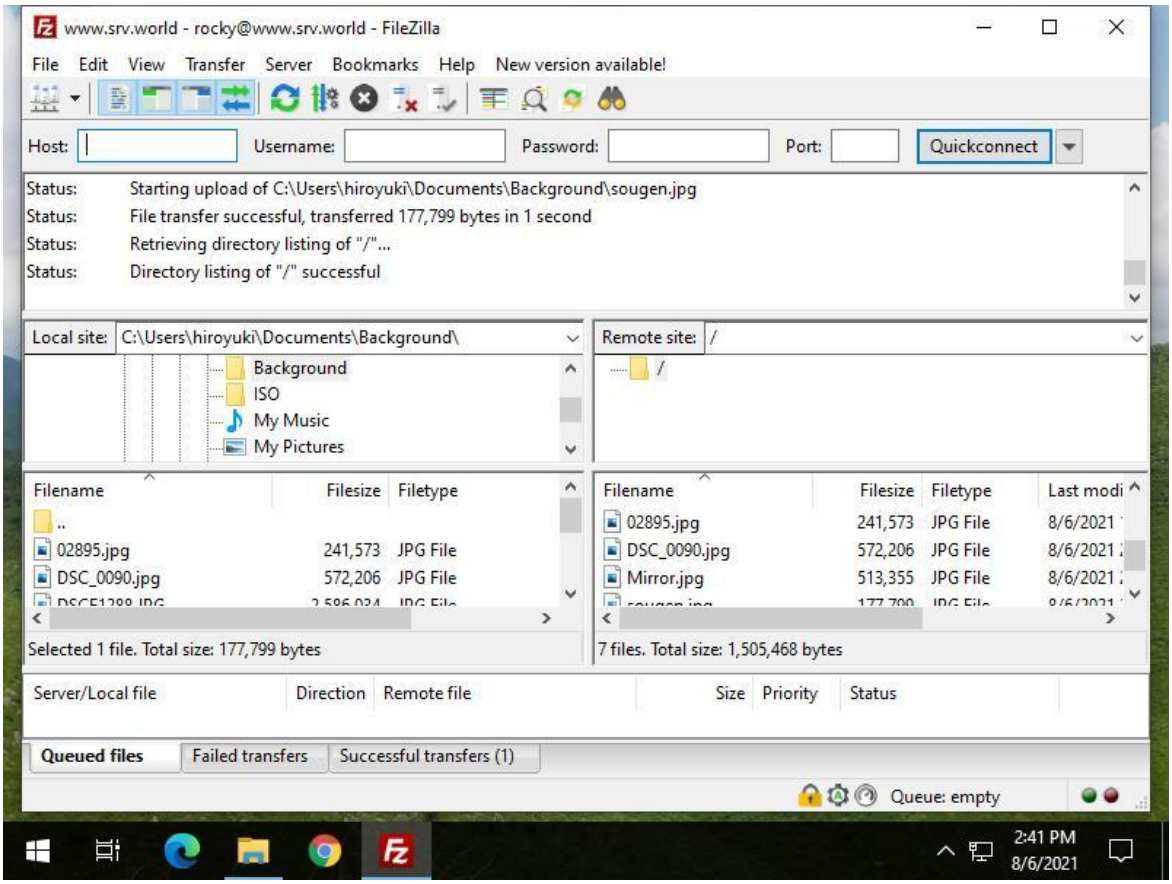
- The main interface will appear.
- Locate the Quickconnect section at the top.



- **Step 2: Connect to the FTP Server**
- Enter the following FTP connection details:

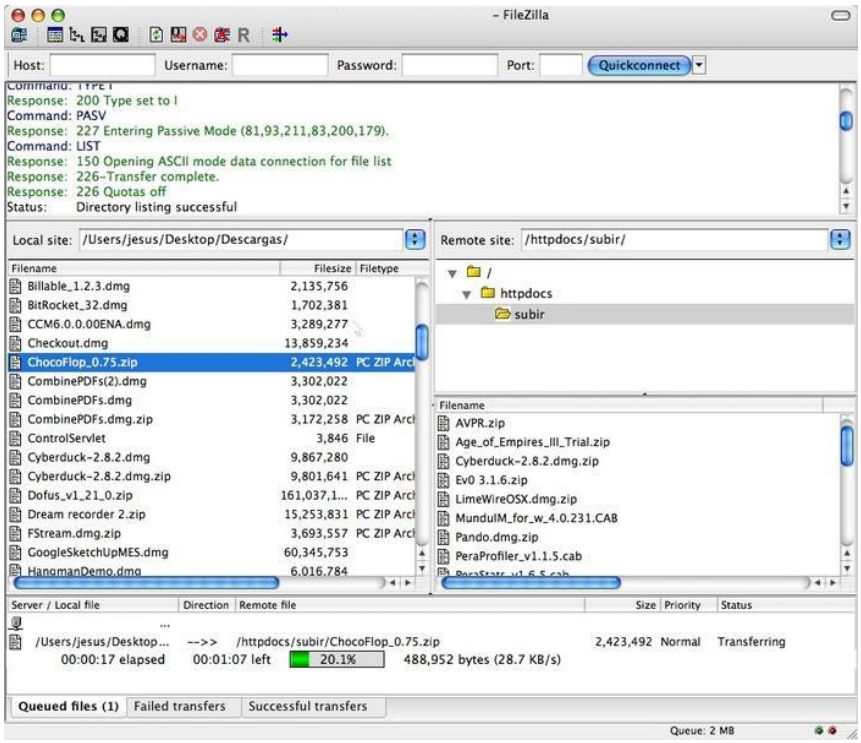
Field	Details
Host	FTP Server IP address
Username	student
Password	Created FTP password
Port	21 (if configured for FTP)

- Enter the server IP address.
- Enter the username and password.
- Click on Quickconnect.
- Wait until the connection is successfully established.
- **Step 3: Locate the Uploaded File**
- On the right side, find the Remote Site section.
- Open the shared folder configured on the FTP server.
- Locate the previously uploaded image file.
- For example, MyPhoto.jpg.

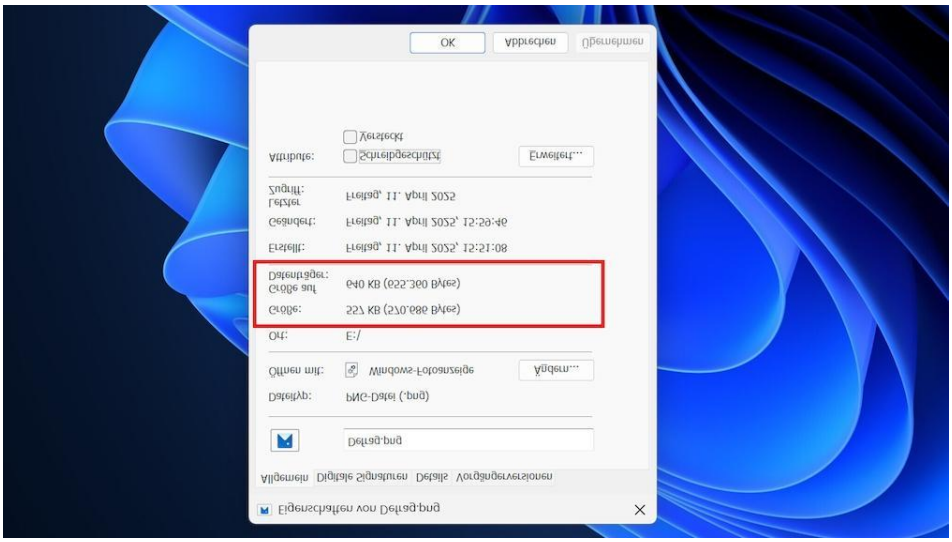


- **Step 4: Create a Different Destination Folder**
- Open File Explorer on your Windows computer.
- Create a new folder in a different location.
- For example, create a folder named FTP_Download.
- This folder will be used to store the downloaded image.
- **Step 5: Download the File Using FileZilla Client**
- In FileZilla Client, locate the Local Site section on the left.
- Browse and open the newly created FTP_Download folder.
- On the right side, select the image file from the Remote Site.
- Right-click the file and select Download.

- Alternatively, drag and drop the file from the Remote Site to the Local Site.
- Wait until the file transfer is completed.



- **Step 6: Verify the Downloaded File**
- Open the FTP_Download folder using File Explorer.
- Locate the downloaded image file.
- Right-click the original image file and select Properties.
- Note its size.
- Now right-click the downloaded image file and select Properties.
- Compare both file sizes.
- If both sizes match, the file has been transferred successfully.



- **File Size Verification**
- The following table shows an example of file size verification:

File Details	File Size
Original Image File	2.5 MB
Downloaded Image File	2.5 MB
Verification	Matching

- **Result**
- Successfully downloaded the previously uploaded image file from the FTP server to a different folder on the local computer using FileZilla Client. The original and downloaded file sizes were compared and found to be matching.

❖ **4. Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites). Think about how network admins monitor device health, bandwidth, or uptime.**

➢ **Aim**

- To identify and explain three real-world scenarios where SNMP (Simple Network Management Protocol) is used in network management to monitor device health, bandwidth usage, network performance, and uptime.

➢ **Introduction**

- **SNMP (Simple Network Management Protocol)** is an Application Layer protocol used to monitor and manage network devices such as routers, switches, servers, printers, and wireless access points.
- It allows network administrators to collect information about device performance, network traffic, CPU usage, memory usage, and device availability.

➢ SNMP generally works through three main components:

1. **SNMP Manager:** The central monitoring system that collects information from network devices.
2. **SNMP Agent:** Software running on a network device that provides information to the SNMP manager.
3. **MIB (Management Information Base):** A collection of managed objects that define the information available from a network device.

➢ SNMP helps administrators identify network problems, monitor device health, and maintain reliable network services.

- **Three Real-World Scenarios Where SNMP Is Used**
- **Scenario 1: Internet Service Providers (ISPs)**
- Internet Service Providers use SNMP to monitor routers, switches, and other networking equipment that provide Internet connectivity to customers.
- **Example:** An ISP providing broadband Internet services to homes and businesses can use SNMP-based monitoring systems to check the health and performance of its network devices.
- **How SNMP is used:**
  - Monitors router and switch CPU and memory usage.
  - Measures incoming and outgoing network bandwidth.
  - Checks whether network interfaces are active or inactive.
  - Monitors device uptime and availability.
  - Helps identify overloaded network links and equipment failures.
- **Real-world benefit:** SNMP helps ISP network administrators detect network problems and maintain reliable Internet connectivity for customers.

- **Scenario 2: College Campus Wi-Fi Networks**
- Colleges and universities use multiple wireless access points, switches, and routers to provide Wi-Fi connectivity to students, teachers, and staff.
- **Example:** A college campus with Wi-Fi available in classrooms, laboratories, libraries, and offices can use SNMP to monitor its networking devices.
- **How SNMP is used:**
  - Monitors the status of wireless access points.

- Checks bandwidth usage across different network devices.
- Tracks device CPU and memory utilization.
- Detects access points that become unavailable.
- Helps administrators identify network congestion.

➢ **Real-world benefit:** SNMP helps college network administrators maintain stable Wi-Fi connectivity and identify problems affecting students and staff.

➢ **Scenario 3: Large Websites and Online Services**

➢ Large websites and online services operate through multiple servers, switches, routers, and data centre networking devices. SNMP can be used to monitor the health and availability of this infrastructure.

➢ **Example:** An online service such as Flipkart may operate a large network infrastructure to support website access, customer requests, and online transactions. In a typical enterprise environment, SNMP-based monitoring can help administrators monitor network devices supporting these services.

➢ **How SNMP is used:**

- Monitors server and network-device availability.
- Tracks network bandwidth and interface traffic.
- Checks CPU and memory utilization on supported devices.
- Monitors device uptime and network interface errors.
- Helps administrators identify performance problems and equipment failures.

➢ **Real-world benefit:** SNMP-based monitoring helps administrators maintain network availability and identify issues that could affect the performance of online services.

➢ Comparison of the Three SNMP Applications

No.	Network Environment	Devices Monitored	Main Purpose
1	Internet Service Provider	Routers and switches	Monitor bandwidth, device health, and uptime
2	College Campus Wi-Fi	Wireless access points, routers, and switches	Monitor Wi-Fi availability and network performance
3	Large Websites	Servers, routers, and switches	Monitor infrastructure health, traffic, and availability

➢ Important Information Monitored Using SNMP

Monitoring Parameter	Purpose
CPU Usage	Checks the processing load on network devices.
Memory Usage	Monitors available and used memory.

Bandwidth Usage	Measures incoming and outgoing network traffic.
Device Uptime	Shows how long a device has been running.
Interface Status	Checks whether a network connection is active or inactive.
Network Errors	Helps identify communication and interface problems.

- **Result**
- Three real-world applications of SNMP were studied: Internet Service Providers, college campus Wi-Fi networks, and large websites or online services. The role of SNMP in monitoring device health, bandwidth usage, network performance, and uptime was understood.

❖ **5. Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.**

➢ **Introduction**

➢ FTP (File Transfer Protocol) and HTTP (Hypertext Transfer Protocol) are two important application-layer protocols used in computer networks.

- FTP: FTP is used to transfer files between a client and a server. It supports uploading and downloading files.
- HTTP: HTTP is used to transfer web pages, images, videos, and other web resources between web browsers and web servers.

➢ **Difference Between FTP and HTTP**

➢ The following table shows the comparison between FTP and HTTP:

Comparison Point	**FTP (File Transfer Protocol)**	**FTP (File Transfer Protocol)**
1. Full Form	File Transfer Protocol	Hypertext Transfer Protocol
2. Port Number	Uses port 21 for control and commonly port 20 for active-mode data transfer.	Uses port 80 by default.
3. Security	Standard FTP does not encrypt data, usernames, or passwords.	Standard HTTP does not encrypt communication. HTTPS provides encryption.

4. Data Transfer Method	Transfers files between a client and server using separate control and data connections.	Transfers web resources using request and response messages.
5. Main Use Cases	Uploading, downloading, and sharing files.	Accessing websites and transferring web content.
6. Authentication	Supports username and password authentication, as well as anonymous access.	Does not require authentication for every request; websites can use passwords, cookies, and tokens.
7. Common Applications	FileZilla, FTP servers, and file-sharing systems.	Web browsers such as Chrome, Firefox, and Edge.